

Data Analysis of Bread Sales

The design of experiments for the bread sales data is a two way factorial fixed effects design. The model for this experiment is:

$$Y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + \varepsilon_{k(ij)}, \quad 1 \leq i \leq 3, 1 \leq j \leq 2, 1 \leq k \leq 2,$$

where the factor A is the height of the shelf, factor B is the width of the shelf, and the response variable is the number of cases of bread sold during the experimental period.

In this experiment we are interested in testing simultaneously the following three null hypotheses:

$$H_{01} : A_i = 0, 1 \leq i \leq 3, \quad (1)$$

$$H_{02} : B_j = 0, 1 \leq j \leq 2, \quad (2)$$

and

$$H_{03} : (AB)_{ij} = 0, 1 \leq i \leq 3, 1 \leq j \leq 2. \quad (3)$$

The test statistic for testing the null hypothesis $H_{01} : A_i = 0, 1 \leq i \leq 3$ is:

$$F_1 = \frac{MS_A}{MS_E},$$

the rest statistic for testing $H_{02} : B_j = 0, 1 \leq j \leq 2$ is:

$$F_2 = \frac{MS_B}{MS_E},$$

and the test statistic for testing $H_{03} : (AB)_{ij} = 0, 1 \leq i \leq 3, 1 \leq j \leq 2$ is:

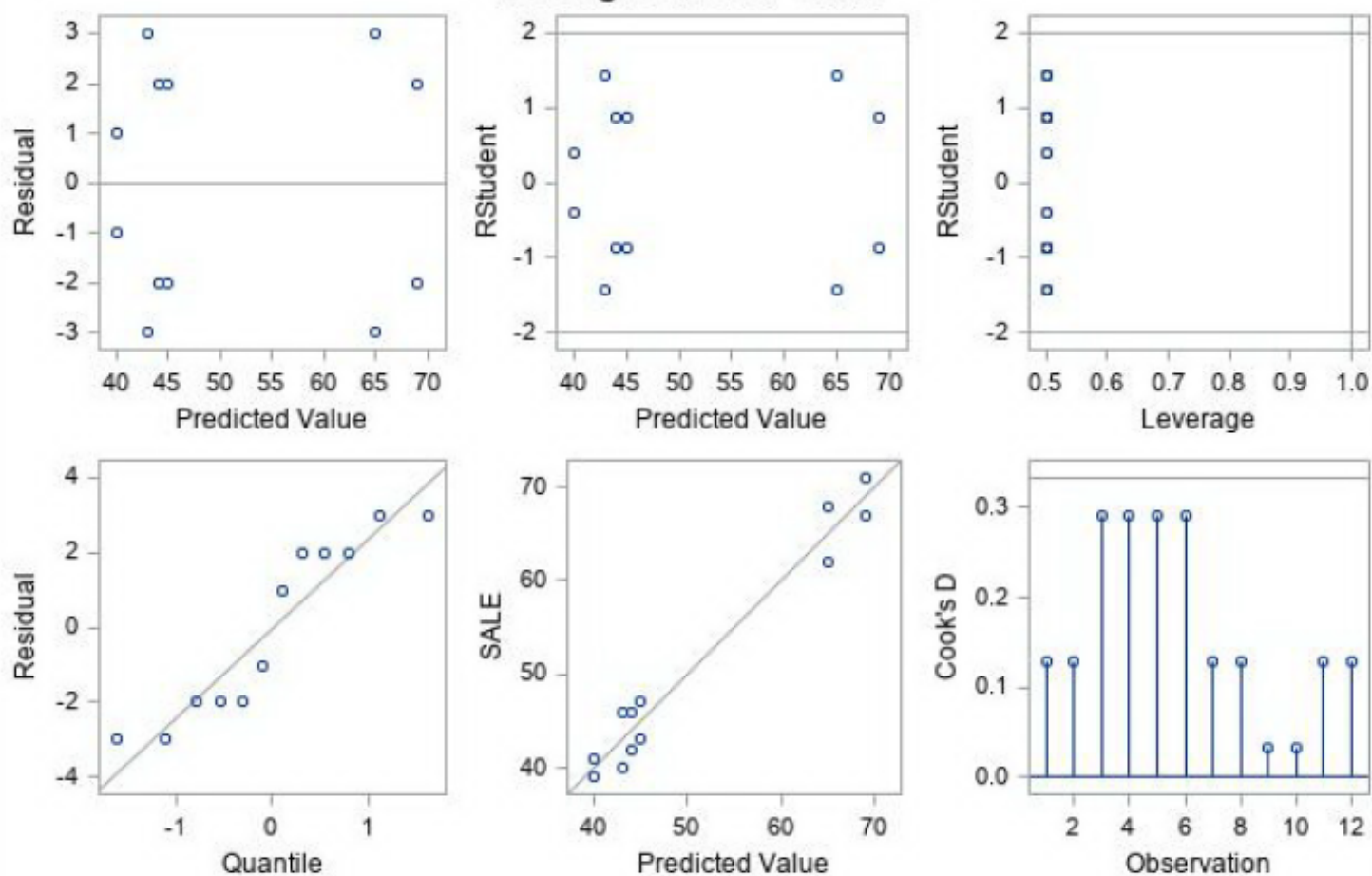
$$F_3 = \frac{MS_{AB}}{MS_E}.$$

Suppose we decide to use an overall significance level $\alpha = .10$. Since we are testing three hypothesis, the nominal alpha level for each test is $\alpha/3 = .10/3 \approx .033$. This means that we are going to reject any of three null hypotheses stated above if the the p-value of their F statics is less than .033. From the SAS output for bread data we get, **if the underlying assumption for our model are valid**, the following table on page 2. Based on this table we get the following.

Since only the F test corresponding to testing the effect of HEIGHT has a p-value $< .0001$, .033, we reject $H_{01} : A_i = 0, 1 \leq i \leq 3$, and conclude that the levels of the shelf height have an effect of the sales of bread. Before we proceed

Source	DF	Type III SS	Mean Square	F Value	Pr > F
HEIGHT	2	1544.000000	772.000000	74.71	<.0001
WIDTH	1	12.000000	12.000000	1.16	0.3226
HEIGHT*WIDTH	2	24.000000	12.000000	1.16	0.3747

Fit Diagnostics for SALE



Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.855765	Pr < W	0.0433

to the multiple comparisons, let us examine the underlying assumptions for our model.

The plot of the Residual vs. Predicted value has no pattern. This implies that linearity of the model and constant variance assumption for the random errors is valid. Based on the plots of the RStudent vs. Predicted value and Cook'sD vs. Observation, we can conclude that the data set has no outliers or influential observations. The normal probability plot, Residual vs. Quantile, has some curvature so we need to consult the p-value of the W statistic.

Since the p-value of the W statistic is .0433, which is less than .10, the normality assumption cannot be confirmed. It is not less than .01, so based on out rule, it cannot be rejected either. Some experimenters will decide to accept the fact that the normality assumption is approximately normal and proceed with the multiple comparisons. Other experimenters will decide to transform the data. Possible transformations include square root transformation of the log transformation. We have discussed in class both transformation, and the related SAS programs and SAS outputs are posted on Husky CT. Suppose the experimenter decided to use the log transformation. Then the following model is considered:

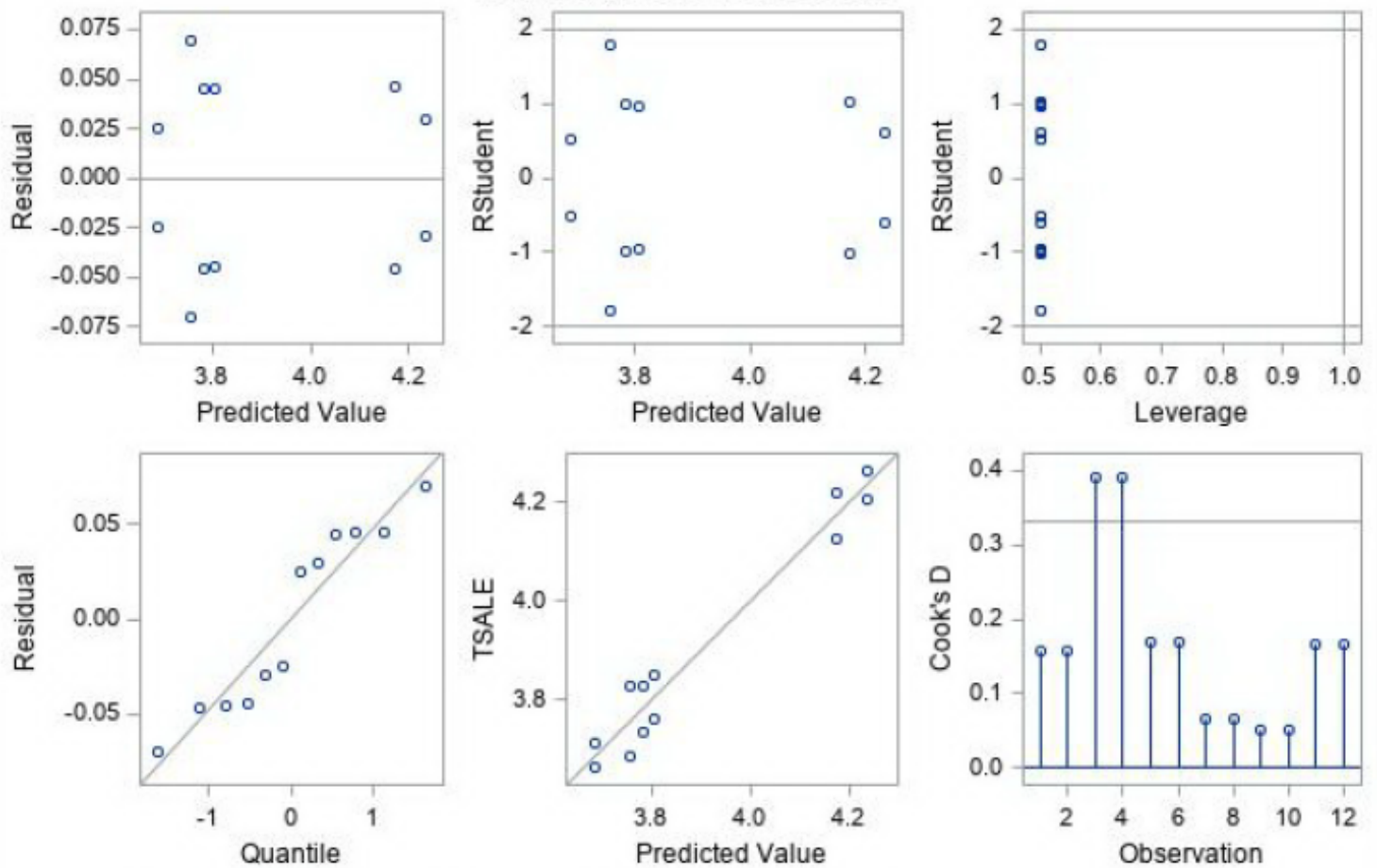
$$Y^* = \log(Y_{ijk}) = \mu + A_i + B_j + (AB)_{ij} + \varepsilon_{k(ij)}, \quad 1 \leq i \leq 3, 1 \leq j \leq 2, 1 \leq k \leq 2.$$

From the SAS output no 2 for bread sales data we get:

Source	DF	Type III SS	Mean Square	F Value	Pr > F
HEIGHT	2	0.53109423	0.26554711	63.58	<.0001
WIDTH	1	0.00389091	0.00389091	0.93	0.3717
HEIGHT*WIDTH	2	0.01090093	0.00545047	1.31	0.3384

and therefore we reject $H_{01} : A_i = 0, 1 \leq i \leq 3$, and conclude that the levels of the shelf height have an effect of the sales of bread. The other two null hypotheses are accepted at the overall $\alpha = .10$ level as the p-value of their F statistics is larger than .033. Let us examine the underlying assumptions for our transformed data model. The Fit Diagnostics plots from the SAS output no 2 for bread sales data are given below, at the end of the file. Based on the Fit Diagnostics plots we can conclude the following. The plot of the Residual vs. Predicted value has no pattern. This implies that linearity of the model and constant variance assumption for the random errors is valid. Based on the plots of the RStudent vs. Predicted value and Cook'sD vs. Observation, we can conclude that the data set has no outliers or influential observations. The normal probability plot, Residual vs. Quantile, has some curvature so we need to consult the p-value of the W statistic. The p-value of the W statistic is .12 .81 > .10, as can be seen in the table given below at the end of this file. Therefore, we confirm the normality assumption for this model and proceed with multiple comparisons for the levels of the height of the shelves. Based on the Tukey multiple comparisons display from SAS output no 2 for bread sales data, given below we can conclude that the average sales of bread for the middle shelf is significantly higher than the average sales for top or bottom shelves. There is no significant difference in average sales for top and bottom shelves. To present an estimate for the average sales for the middle height shelf, we need to transform the value to the original scale: $\exp(4.2036) = 66.92$. The average sales for lower shelf $\exp(3.7359)=41.93$, and for the upper shelf the average number of cases sold is $\exp(3.7822)=43.91$.

Fit Diagnostics for TSALE



Tests for Normality

Test	Statistic		p Value	
Shapiro-Wilk	W	0.892822	Pr < W	0.1281

TSALE Tukey Grouping for Means of HEIGHT (Alpha = 0.05)

Means covered by the same bar are not significantly different.

HEIGHT	Estimate
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2	4.2035
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1	3.7822
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3	3.7359
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